

Jayacharan R

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Education

Vellore Institute Of Technology, *B.Tech in Computer Science (Data Science) – Vellore* | **CGPA: 9.27** July 2023 – present
Achyuta Public School, *CBSE, Highschool in Bio-Mat – Dindigul, TN* | **AISSE: 89.6%, AISSE: 98%** June 2017 – Apr 2023

Projects

WikiMind: End-to-End Agentic RAG & Knowledge Engine June 2026 – present

- Architected the pipeline using LangGraph to orchestrate two-stage local retrieval (dense and sparse) over a Qdrant vector database, integrating NeMo Guardrails for LLM middleware and Langfuse for observability and telemetry.
- Engineered advanced retrieval modalities including entity-based Knowledge Graph traversal (spaCy/NetworkX), temporal versioning for time-travel queries, and automated evaluation harnesses measuring Recall@K/MRR, while extending offline inference to mobile environments utilizing ONNX-quantized embeddings and SQLite-vss.

Clean-Backup: Intelligent Media Automation & Cloud Sync Engine Jan 2026 – present

- Architected a hybrid Python and Rust media organization engine featuring rapid perceptual image hashing (pHash) for deduplication, and integrated the Google Drive API via OAuth 2.0 for automated cloud synchronization.
- Engineered a high-throughput, multi-core media compression pipeline and orchestrated asynchronous background jobs, paired with a transactional journal system ensuring atomic file operations and complete state rollback.

Experience

AI-ML Intern, *Tata Consultancy Services - TCS – Chennai, TN, India* June 2026 – July 2026

- Architected a secure RAG pipeline using FastAPI and LangChain, implementing custom Aegis guardrails to enforce contextual relevance and mitigate prompt injection risks.
- Established full-stack LLM observability with real-time dashboards, instrumenting execution traces to monitor retrieval latency, token utilization, and generation quality.

SAP Analytics Cloud Intern, *Blueprint Technologies Pvt Ltd – Bengaluru, KA, India* May 2025 – July 2025

- Developed responsive dashboards in SAP Analytics Cloud by conducting exploratory data analysis, cleaning data, and integrating live Google Drive feeds with time-based metrics to enable real-time performance monitoring.
- Implemented predictive models (linear regression, classification, time series) and advanced dimension-level calculations to deepen analytical insights, while successfully resolving geo-mapping issues by debugging latitude-longitude ID conflicts.

Awards

Winner - Roboverse Ideathon 2025, **IEEE Robotics and Automation Society, VIT, Vellore** Jan 2025

- Engineered an AI-driven surveillance prototype using YOLO (Ultralytics) for real-time anomaly detection and human movement tracking in CCTV footage. Demonstrated crime prevention capabilities by implementing automated monitoring, behavioral analysis, and security alerts with comprehensive event logging.

Skills

Programming & Frameworks: Python, Java, C, C++, Bash, YAML, HTML, CSS, JS, Pillow, FFmpeg, Tkinter

AI/ML Frameworks: PyTorch, LangChain, Langfuse, RAG, LLM Observability, LLM Guardrails, Quantization, Fine-tuning, HuggingFace Embeddings, ONNX Runtime, Qdrant (Vector DB), Redis, NetworkX (Knowledge Graphs), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, NVIDIA NeMo Guardrails

Software: SAP Analytics Cloud, Git Version Control, AWS Architecture, Docker Deployment, Arch Linux

Language: English (Professional proficiency), Tamil (Native proficiency), Japanese & Hindi (Limited proficiency)

Certifications

AWS Certified Solutions Architect – Associate Dec 2025 – Dec 2028

Microsoft Certified: Azure AI Engineer Associate - AI 102 Nov 2025 – Nov 2026

Microsoft Certified: Azure Data Fundamentals - DP 900 Oct 2025